

Onondaga County (Syracuse) Sample — Shopping Center / Retail



Traffic Impact Study

Prepared August 18, 2026 · 43.0500, -76.0800

SIS sample draft — full tier

Traffic Impact Study

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TRAFFIC IMPACT STUDY FOR DOT PROJECTS

Prepared in Accordance With Chapter 5 of the NYSDOT Highway Design Manual (HDM)

Project name	Onondaga County (Syracuse) Sample — Shopping Center / Retail
PIN	PIN xxxx.xx (to be assigned by NYSDOT)
Project description	Shopping Center / Retail (85 1,000 sqft GLA) within Syracuse MSA.
NYSDOT Region	Region 3 — Syracuse / Central New York
Site coordinates	43.0500°, -76.0800°
Opening year (ETC)	2027

Note:

It is a violation of law for any person, unless they are acting under the direction of a licensed professional engineer, architect, landscape architect, or land surveyor, to alter an item in any way. If an item bearing the stamp of a licensed professional is altered, the altering engineer, architect, landscape architect, or land surveyor shall stamp the document and include the notation "altered by" followed by their signature, the date of such alteration, and a specific description of the alteration.

This report is based on the NYSDOT TIS Shell revised on 9/16/2014.

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PROJECT CONTEXT

NYSDOT region: Region 3 — Syracuse / Central New York

Planning group: SMTC (Syracuse Metropolitan Transportation Council)

Regulatory regime: SEQRA (ECL Article 8 / 6 NYCRR Part 617).

SEQRA classification: Not yet determined — lead agency to classify as Type I (6 NYCRR Part 617.4 — likely EIS), Type II (Part 617.5 — exempt), or Unlisted (Part 617.6 — Short EAF + significance finding) before locking the TIS.

1.0 SUMMARY OF TRAFFIC IMPACTS

Capacity:

A capacity analysis was performed per NYSDOT Highway Design Manual Chapter 5. The following capacity improvement measures are cost effective, and are recommended to reduce reoccurring delay: Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s; Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s.

Crash:

A minimum three-year accident history review was performed and the safety screening met all of the required steps. A full crash analysis is not required.

GML §239-m / §239-n county planning board referral required:

The site is within 500 ft of THOMPSON RD (Urban Minor Arterial), which triggers mandatory referral to the controlling county planning board under General Municipal Law §239-m (site-plan / special-permit / variance / zoning amendment) or §239-n (subdivision review) before the local board may act. The county may make adverse findings under §239-f; the local board can override only by supermajority. Practical TIS consequence: the county will typically request the TIS in full as part of the §239 referral package — pre-application coordination with the county planning agency is recommended.

INTERSECTIONS

2

LOS DROPS

AT LOS E/F

14.9s

WORST Δ DELAY

2.0 TRAVEL SPEEDS

Per NYSDOT HDM Chapter 5 §5.2, posted speed limits and actual operating speeds (85th-percentile) along the study network are reported below. Posted-speed-limit values in the third column are auto-ingested from the NYSDOT Roadway Data Management (RDM) FeatureServer (https://gis.dot.ny.gov/hostingny/rest/services/Roadways/RDM_Roadway_Current/FeatureServer) at PDF-generation time — the same source data viewable through the NYSDOT Traffic Data Viewer (<https://gis.dot.ny.gov/html5viewer/?viewer=tdv>). Operating speeds (85th-percentile) are NOT published by NYSDOT and require a project-specific radar / floating-car speed study.

Street	Functional class	Posted speed	85th-pctile operating speed
Near Erie Boulevard East (ERIE BLVD E NY 5)	Urban Principal Arterial Other	40 mph	Speed study required
Thompson Road (THOMPSON RD)	Urban Minor Arterial	30 mph	Speed study required
Erie Boulevard East (Bridge St)	Urban Principal Arterial Other	40 mph	Speed study required
Near Bridge Street (Bridge St)	Urban Principal Arterial Other	40 mph	Speed study required
Near Erie Boulevard East	Urban Principal Arterial Other	40 mph	Speed study required
Near Bridge Street (WIDEWATERS PKWY)	Urban Local	30 mph	Speed study required

3.0 CAPACITY ANALYSIS

Capacity analyses performed in this report are consistent with HCM 6th Edition. The software used to perform this analysis is the SimpleImpactStudies HCM 6 solver (per-approach control-delay model). Synchro / SimTraffic, HCS, Sidra, or VISSIM may be substituted for formal submittal per NYSDOT Regional Traffic Office preference.

Study network functional-class composition (per NYSDOT RDM): Urban Principal Arterial Other (7); Urban Minor Arterial (1); Urban Local (1). NYSDOT functional class drives access management expectations under HDM Appendix 5A (entrance standards for state highways) and the default K-factor banding under HDM §5.3.

Level of Service thresholds applied below are per HCM 2010 Exhibits 18-4 (signalized), 19-1 / 20-2 / 21-1 (unsignalized — 2-way stop / all-way stop / roundabout), and 10-7 (freeway). Any component $v/c > 1.0$ is LOS F regardless of delay or density.

Signalized intersections (HCM 2010 Exhibit 18-4, p. 18-6) — control delay sec/veh; any $v/c > 1.0$ is F.

LOS	A	B	C	D	E	F
sec/veh	≤ 10	10-20	20-35	35-55	55-80	> 80

Unsignalized — 2-way stop / all-way stop / roundabout (HCM 2010 Exhibits 19-1, 20-2, 21-1) — sec/veh; any $v/c > 1.0$ is F.

LOS	A	B	C	D	E	F
sec/veh	≤ 10	10-15	15-25	25-35	35-50	> 50

Freeway basic / weaving / merge-diverge (HCM 2010 Exhibit 10-7, p. 10-9) — density pc/mi/ln; any component $vd/c > 1.00$ is F.

LOS	A	B	C	D	E	F
pc/mi/ln	≤ 11	11-18	18-26	26-35	35-45	> 45

3.1 Growth Rates

Background traffic growth is applied at 1.5% per year over 1 year. This rate is the per-station median CAGR measured from the NYSDOT Traffic Monitoring AADT FeatureServer for Region 3 — Syracuse / Central New York: 0.00%/yr median across 1,492 count stations (rolling actual-count window approximately 2000-2019; 25th-75th percentile range -1.16% to 1.39%). The measured value supersedes the NYSDOT statewide default fallbacks (per HDM Chapter 5: 1.7%/yr Interstate + ramps from the Data Services Bureau, 2.0%/yr all other roads from the Regional Planning Group). The

controlling Regional Planning Group SMTC (Syracuse Metropolitan Transportation Council) may have separately-adopted growth-rate guidance that overrides the measured median for formal submittal — confirm at scoping.

3.2 Existing AADT and DHV

Existing AADT is sourced from the NYSDOT Traffic Data Viewer. Design Hourly Volume (DHV) is reported per HDM Chapter 5 convention as AADT × K-factor; this analysis applies K = 0.10 as the statewide design default — facility-specific K should be substituted from NYSDOT count-station records for formal submittal. The table below grows the existing volume at 1.5%/yr to Estimated Time of Completion (ETC), ETC + 20 years, and ETC + 30 years per HDM Chapter 5 §5.3.

Intersection	2026 AADT	2026 DHV	2027 (ETC) AADT	2027 (ETC) DHV	2047 AADT	2047 DHV	2057 AADT	2057 DHV
Near Erie Boulevard East	18,820	1,882	19,102	1,910	25,728	2,573	29,858	2,986
Thompson Road & Erie Boulevard East	5,480	548	5,562	556	7,491	749	8,694	869
Erie Boulevard East & Bridge Street	18,820	1,882	19,102	1,910	25,728	2,573	29,858	2,986
Near Bridge Street	7,080	708	7,186	719	9,679	968	11,233	1,123
Near Erie Boulevard East	8,210	821	8,333	833	11,224	1,122	13,025	1,303
Near Bridge Street	7,079	708	7,185	719	9,677	968	11,231	1,123
Near Erie Boulevard East	1,000	100	1,015	101	1,367	137	1,587	159
Near East Genesee Street	999	100	1,014	101	1,366	137	1,585	158
Near Bridge Street	7,080	708	7,186	719	9,679	968	11,233	1,123

Source: AADT estimated from engine peak-hour entering-volume sum via K = 0.10. NYSDOT Traffic Data Viewer measured AADT is recommended for submittal. Projection applies the 1.5%/yr background growth rate compounded annually.

3.3 Traffic Control Device Data

Existing traffic control devices at each study-network intersection are summarized below. Signal timing data (cycle length, phase splits, offset) and per-approach detection presence should be confirmed against the controlling NYSDOT Regional Traffic Office or municipal traffic-engineering signal record for formal submittal.

Intersection	Control type	Approaches w/ detection
Near Erie Boulevard East	Signal (controlling RTO to confirm)	Field verification required
Thompson Road & Erie Boulevard East	Signal (controlling RTO to confirm)	Field verification required
Erie Boulevard East & Bridge Street	Signal (controlling RTO to confirm)	Field verification required
Near Bridge Street	Signal (controlling RTO to confirm)	Field verification required
Near Erie Boulevard East	Signal (controlling RTO to confirm)	Field verification required
Near Bridge Street	Signal (controlling RTO to confirm)	Field verification required
Near Erie Boulevard East	Signal (controlling RTO to confirm)	Field verification required
Near East Genesee Street	Signal (controlling RTO to confirm)	Field verification required
Near Bridge Street	Signal (controlling RTO to confirm)	Field verification required

3.4 Capacity Analysis for Existing No-Build Condition

Existing No-Build conditions reflect current-year peak-hour volumes (no growth, no project trips). Existing measured volumes are grown to opening year 2027 at 1.5%/yr to establish the No-Build baseline for §3.5 comparison.

Intersection	Existing LOS	Existing delay (s)	No-Build LOS	No-Build delay (s)	v/c
Near Erie Boulevard East	E	68.7	E	73.7	1.06

Intersection	Existing LOS	Existing delay (s)	No-Build LOS	No-Build delay (s)	v/c
Thompson Road & Erie Boulevard East	B	16.7	B	16.8	0.31
Erie Boulevard East & Bridge Street	E	68.7	E	73.7	1.06
Near Bridge Street	B	18.0	B	18.1	0.40
Near Erie Boulevard East	B	19.0	B	19.1	0.46
Near Bridge Street	B	18.0	B	18.1	0.40
Near Erie Boulevard East	B	14.1	B	14.1	0.06
Near East Genesee Street	B	14.1	B	14.1	0.06
Near Bridge Street	B	18.0	B	18.1	0.40

Trip Distribution by Time of Day

Estimated within-day distribution of the 6,800 gross daily trips and the resulting on-site accumulation. Within-day distribution from NHTS 2022 shopping / errand / meal trip start times, cross-checked against public-data time-of-day patterns for LU 820 / 850. Activity builds to a midday plateau and a stronger late-afternoon/evening peak; on-site occupancy peaks midday through the evening.

Figure — Inbound / Outbound Trips by Start Time

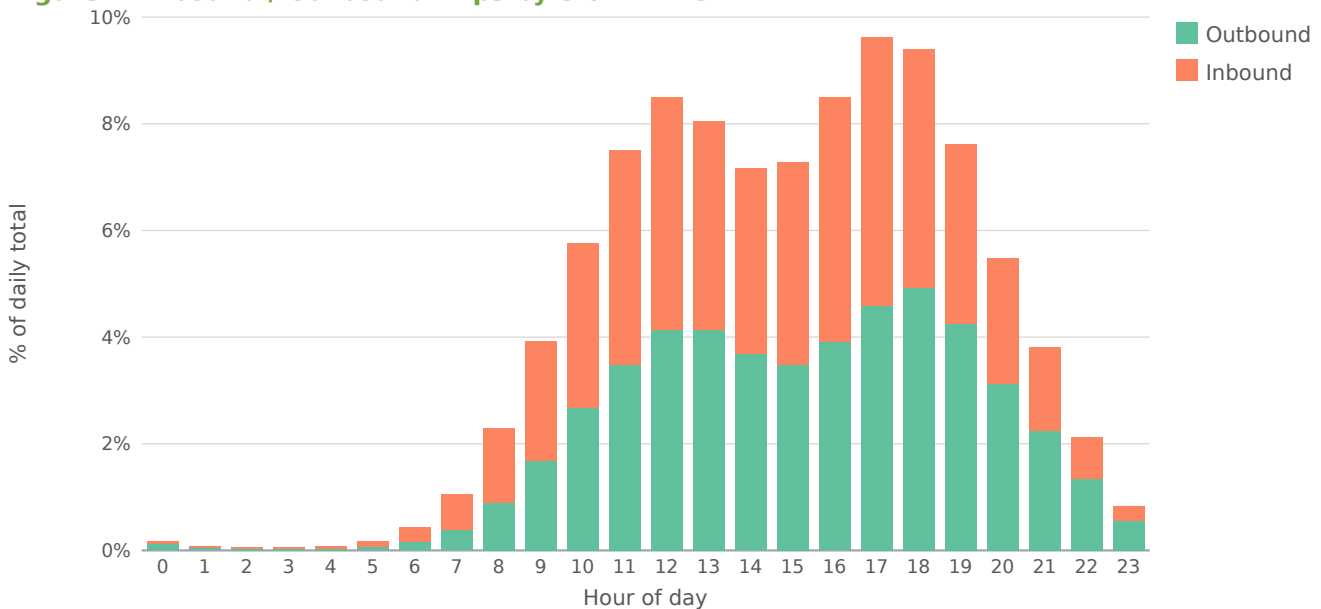
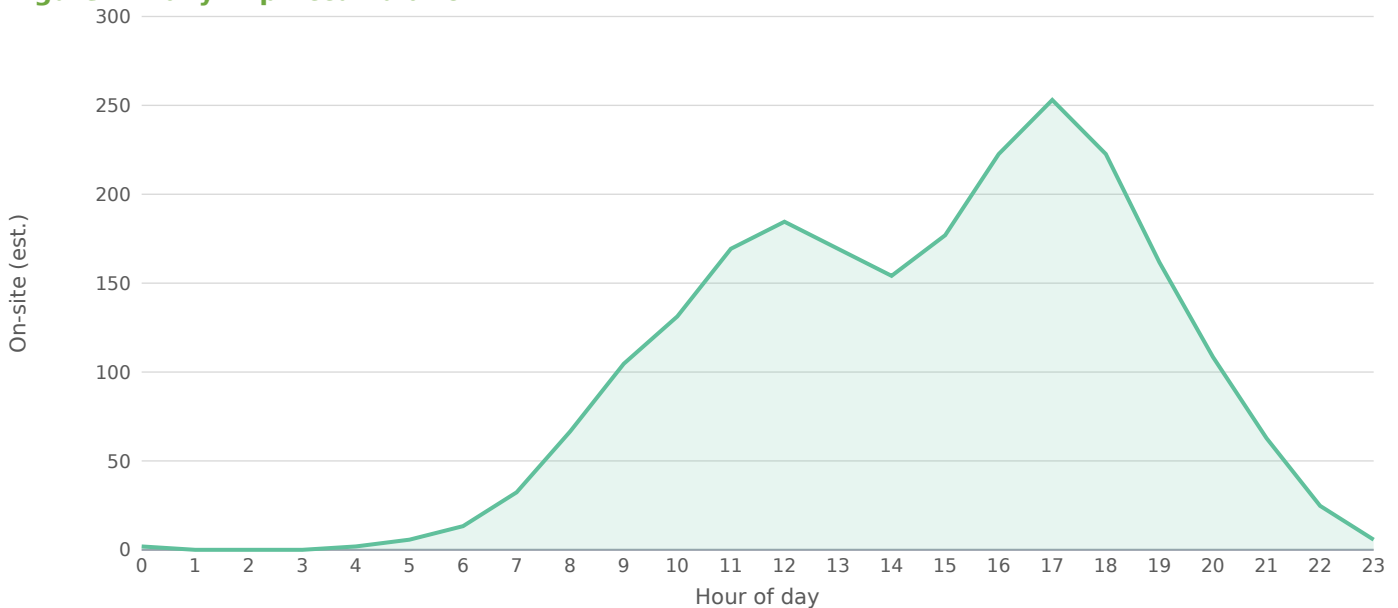


Figure — Daily Trip Accumulation



Peak on-site accumulation ~253 at 17:00, from 6,800 gross daily trips on the retail within-day profile. Screening estimate; not a substitute for a calibrated time-of-day model.

3.5 Capacity Analysis for Proposed Build Condition

Build conditions add the project's external peak-hour trips (680 PM peak, 326 inbound / 354 outbound) to the No-Build baseline at each affected intersection. Per HDM Chapter 5, intersection projects should also analyze each adjacent intersection plus any signal within ½ mile of the project site if expected left-turn volume changes materially affect capacity. The engine's network is determined by a 0.75-mile screening radius; manual scoping should confirm the ½-mile left-turn rule is satisfied.

Intersection	ETC No-Build	ETC Build	ETC+20 NB	ETC+20 Bld	Δ delay (s)
Near Erie Boulevard East	E	▲ F	F	F	14.9
Thompson Road & Erie Boulevard	B	B	B	B	0.4
East Erie Boulevard East & Bridge Street	E	▲ F	F	F	10.2
Near Bridge Street	B	B	C	C	0.3
Near Erie Boulevard East	B	B	C	C	0.3
Near Bridge Street	B	B	C	C	0.1
Near Erie Boulevard East	B	B	B	B	0.1
Near East Genesee Street	B	B	B	B	0.1
Near Bridge Street	B	B	C	C	0.1

3.6 Capacity Improvement Measures

The following capacity improvements are recommended at affected intersections under Build conditions. Improvement category guidance per HDM Chapter 5: addition of turn lanes; installation or re-timing of traffic signals; signage and striping to prevent unwanted turning movements; conversion to roundabout; and TDM / TSM measures per HDM Chapter 24.

Near Erie Boulevard East [MAJOR]

Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s.

Erie Boulevard East & Bridge Street [MAJOR]

Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s.

3.7 Trip Distribution — Gravity Model

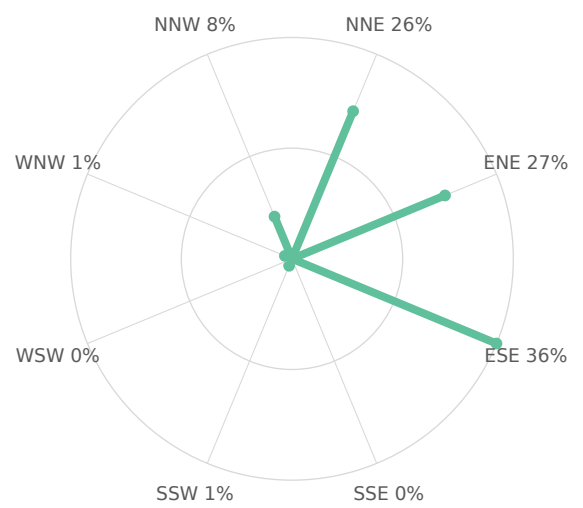
The trip distribution uses the gravity model ($\text{term} = M / (d^1 \cdot d_{\text{site}})$); mass basis: intersection through-volume (AADT $\times$ K-factor) as the destination-activity attraction proxy). Shares are normalized to 100% of project trips and drive the trip assignment below. Basis: NCHRP-716 gamma-friction gravity model (production-constrained) over study-area attraction zones.

Directional sector		Share of project trips			
Northeast (NNE + ENE)		54%			
Southeast (ESE + SSE)		36%			
Southwest (SSW + WSW)		1%			
Northwest (WNW + NNW)		9%			

Study-area zone	Dir.	Distance (mi)	Mass (M)	Gravity term	Trip share
Near Erie Boulevard East	NNE	0.35	1,882	0.3	26.29%
Erie Boulevard East & Bridge Street	ESE	0.42	1,882	0.3	25.64%
Near Erie Boulevard East	ESE	0.54	821	0.1	10.78%
Near Bridge Street	ENE	0.50	708	0.1	9.39%
Near Bridge Street	ENE	0.61	708	0.1	9.07%
Near Bridge Street	ENE	0.71	708	0.1	8.79%
Thompson Road & Erie Boulevard East	NNW	0.41	548	0.1	7.52%
Near Erie Boulevard East	WNW	0.64	100	0.0	1.27%
Near East Genesee Street	SSW	0.71	100	0.0	1.24%

9 study-area zone(s). Screening-grade distribution; not a calibrated regional model.

Figure — Directional Distribution of Project Trips



Screening-grade directional distribution of net new project trips by compass octant (spoke length $\propto$ percent of project trips). Derived from the Gravity Model distribution.

Figure — Project Trip Share by Study-Area Zone

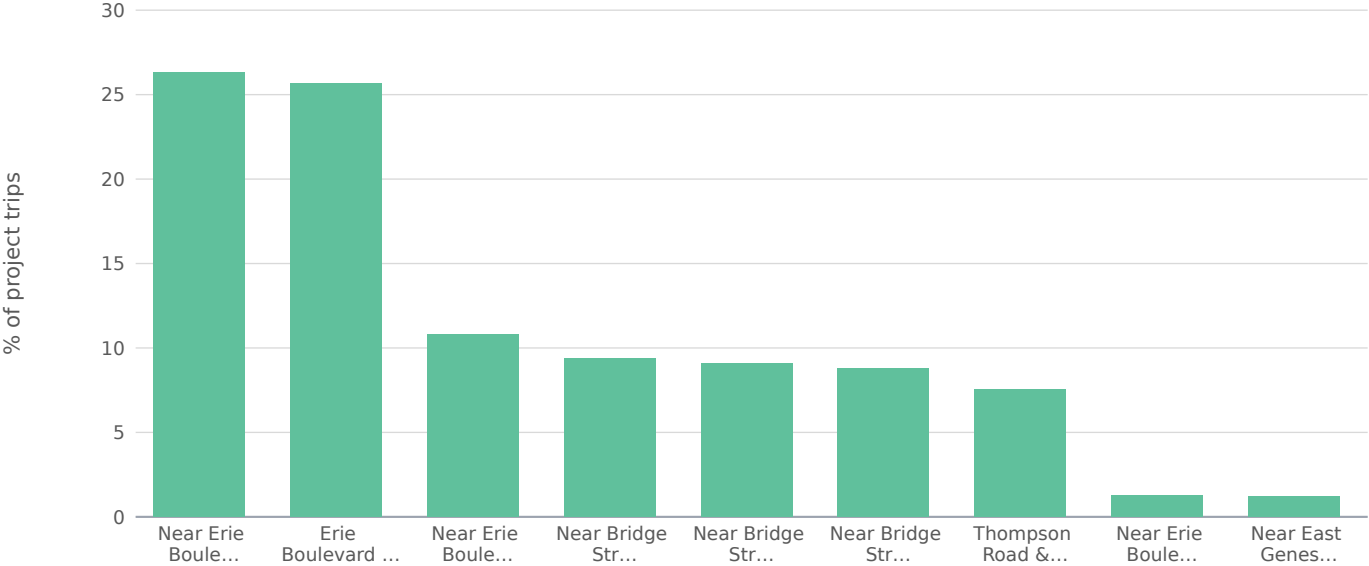
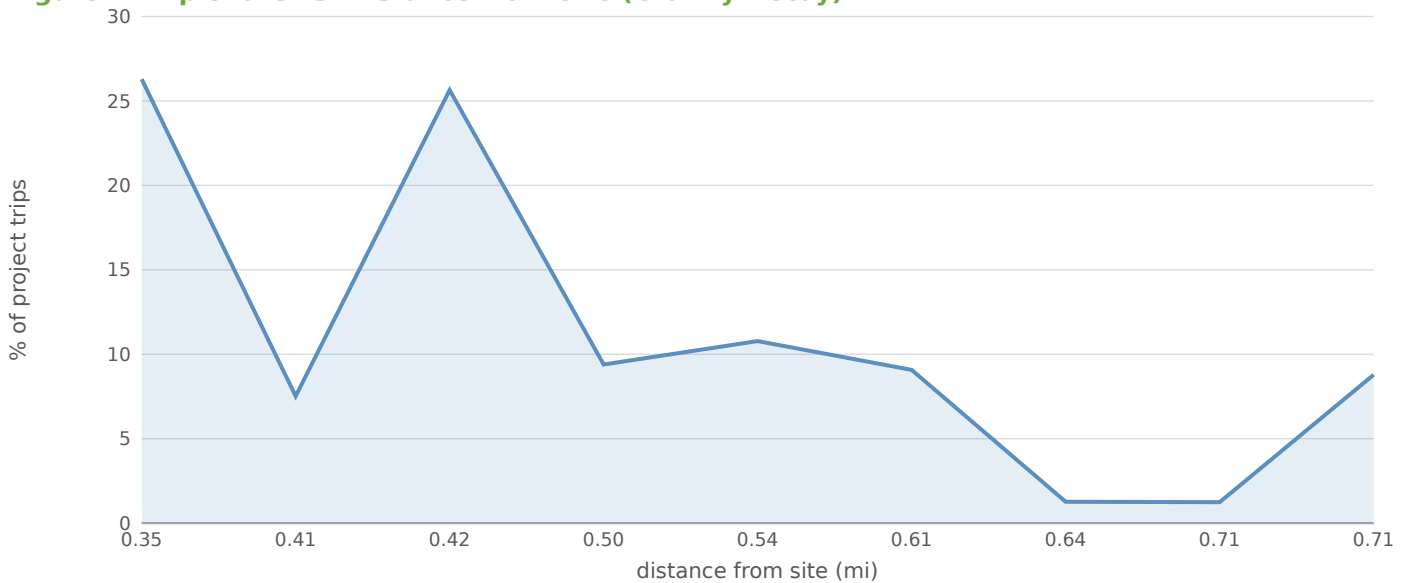


Figure — Trip Share vs. Distance from Site (Gravity Decay)



3.8 Project Trip Assignment

Study intersection	Dir.	Distance (mi)	AM trips	PM trips	Share of project
Near Erie Boulevard East	NNE	0.35	41	77	23.9%
Thompson Road & Erie Boulevard East	NNW	0.41	31	59	18.3%
Erie Boulevard East & Bridge Street	ESE	0.42	28	54	16.8%
Near Bridge Street	ENE	0.50	19	36	11.2%
Near Erie Boulevard East	ESE	0.54	16	31	9.6%
Near Bridge Street	ENE	0.61	11	21	6.5%
Near Erie Boulevard East	WNW	0.64	9	18	5.6%
Near East Genesee Street	SSW	0.71	7	13	4.0%
Near Bridge Street	ENE	0.71	7	13	4.0%

Project trips assigned to each study intersection from the distribution shares above: the directional shares orient the loading toward the high-share sectors, distance-decayed to each approach.

4.0 CRASH ANALYSIS

A minimum three-year accident history review was performed for the period 2023 to 2026 per HDM Chapter 5 §5.3.4. County-level crash counts below are auto-ingested from the New York State Police "Motor Vehicle Crashes – Case Information: Three Year Window" Open Data dataset (<https://data.ny.gov/Transportation/Motor-Vehicle-Crashes-Case-Information-Three-Year-/e8ky-4vqe>), filtered by county_name resolved from the project site coordinates via the NYSDOT RDM_Roadway_Current FeatureServer. The values cover the entire rolling 3-year reporting window for ONONDAGA County, statewide-system-wide — they are NOT a per-segment or per-intersection rate.

Severity descriptor (NY DMV)	Count (3-yr rolling)	Annualized
Fatal Accident	108	36.0

Severity descriptor (NY DMV)	Count (3-yr rolling)	Annualized
Injury Accident	1,023	341.0
Property Damage Accident	37,403	12467.7
Property Damage & Injury Accident	8,062	2687.3
Total (all severities)	46,596	15532.0

Within ONONDAGA County over the rolling 3-year window, 46,596 crashes were reported to NY DMV (108 fatal, 9,085 injury-involved, 37,403 property damage only). The Fatal share is 0.077% of all reported crashes (annualized). For the Safety Screening per HDM Chapter 5 §5.3.4, the project-specific rate must be normalized against the controlling facility's exposure (segment AADT × segment length; intersection AADT × entry-flow) for comparison to NYSDOT Office of Modal Safety statewide averages and the 1.5× HAL trigger. County-totals above provide a county-baseline reference only — site-radius (1/4-mile typical) crash records, HAL / PIL / SDL flags, and CMF / CRF mitigation values must be obtained from the Region 3 — Syracuse / Central New York Regional Traffic Office via FOIL or SIMS. Required NYSDOT forms when the §4.0 section is expanded for formal submittal: TE-156a (Collision Diagram), TE-164a (Safety Benefits Evaluation), TE-204a (Accident Rate Summary), TE-213 (HAL / PIL / SDL Screening Output).

APPENDIX A — EXISTING VOLUME REPORT

Per-approach existing peak-hour volumes (vph) for all affected intersections within study limits.
Source: SimpleImpactStudies HCM 6 solver, calibrated against the controlling DOT 511 / Regional Traffic Office data feed.

Intersection	Approach	Existing (vph)	No-Build (vph)	Build (vph)
Near Erie Boulevard East	NB	557	563	605
Near Erie Boulevard East	SB	472	479	497
Near Erie Boulevard East	EB	469	476	476
Near Erie Boulevard East	WB	384	390	409
Thompson Road & Erie Boulevard East	NB	160	162	193
Thompson Road & Erie Boulevard East	SB	141	143	171
Thompson Road & Erie Boulevard East	EB	129	131	132
Thompson Road & Erie Boulevard East	WB	118	120	120
Erie Boulevard East & Bridge Street	NB	637	647	647
Erie Boulevard East & Bridge Street	SB	402	408	412
Erie Boulevard East & Bridge Street	EB	492	499	527
Erie Boulevard East & Bridge Street	WB	352	357	379
Near Bridge Street	NB	223	226	226
Near Bridge Street	SB	184	186	190
Near Bridge Street	EB	164	167	186
Near Bridge Street	WB	137	140	153
Near Erie Boulevard East	NB	271	275	275
Near Erie Boulevard East	SB	212	215	215
Near Erie Boulevard East	EB	186	189	205
Near Erie Boulevard East	WB	152	155	169
Near Bridge Street	NB	190	193	193
Near Bridge Street	SB	181	184	186
Near Bridge Street	EB	182	185	196
Near Bridge Street	WB	155	158	165
Near Erie Boulevard East	NB	31	31	31
Near Erie Boulevard East	SB	21	21	29
Near Erie Boulevard East	EB	27	28	29
Near Erie Boulevard East	WB	21	22	31
Near East Genesee Street	NB	28	29	34
Near East Genesee Street	SB	25	25	32
Near East Genesee Street	EB	23	24	25
Near East Genesee Street	WB	23	24	24
Near Bridge Street	NB	192	195	195
Near Bridge Street	SB	185	188	189
Near Bridge Street	EB	184	186	193

Intersection	Approach	Existing (vph)	No-Build (vph)	Build (vph)
Near Bridge Street	WB	148	150	154

APPENDIX B — EXISTING CONDITION CAPACITY ANALYSIS OUTPUT

Per-intersection HCM 6 solver output for the Existing No-Build condition (current-year volumes; volumes grown to opening year at the §3.1 background-growth rate for No-Build comparison).

Intersection	Approach	Vol (vph)	v/c	Delay (s)	LOS
Near Erie Boulevard East	NB	565	0.70	24.8	C
Near Erie Boulevard East	SB	479	0.59	21.7	C
Near Erie Boulevard East	EB	476	0.59	21.6	C
Near Erie Boulevard East	WB	390	0.48	19.4	B
Thompson Road & Erie Boulevard	NB	162	0.20	15.5	B
East Thompson Road & Erie Boulevard	SB	143	0.18	15.3	B
East Thompson Road & Erie Boulevard	EB	131	0.16	15.1	B
East Thompson Road & Erie Boulevard	WB	120	0.15	15.0	B
East Erie Boulevard East & Bridge Street	NB	647	0.80	29.3	C
Erie Boulevard East & Bridge Street	SB	408	0.50	19.8	B
Erie Boulevard East & Bridge Street	EB	499	0.62	22.3	C
Erie Boulevard East & Bridge Street	WB	357	0.44	18.7	B
Near Bridge Street	NB	226	0.28	16.4	B
Near Bridge Street	SB	186	0.23	15.8	B
Near Bridge Street	EB	167	0.21	15.6	B
Near Bridge Street	WB	140	0.17	15.2	B
Near Erie Boulevard East	NB	275	0.34	17.2	B
Near Erie Boulevard East	SB	215	0.27	16.3	B
Near Erie Boulevard East	EB	189	0.23	15.9	B
Near Erie Boulevard East	WB	155	0.19	15.4	B
Near Bridge Street	NB	193	0.24	15.9	B
Near Bridge Street	SB	184	0.23	15.8	B
Near Bridge Street	EB	185	0.23	15.8	B
Near Bridge Street	WB	158	0.19	15.5	B
Near Erie Boulevard East	NB	31	0.04	13.9	B
Near Erie Boulevard East	SB	21	0.03	13.8	B
Near Erie Boulevard East	EB	28	0.03	13.9	B
Near Erie Boulevard East	WB	22	0.03	13.8	B
Near East Genesee Street	NB	29	0.04	13.9	B
Near East Genesee Street	SB	25	0.03	13.9	B
Near East Genesee Street	EB	24	0.03	13.9	B
Near East Genesee Street	WB	24	0.03	13.9	B
Near Bridge Street	NB	195	0.24	16.0	B
Near Bridge Street	SB	188	0.23	15.9	B
Near Bridge Street	EB	186	0.23	15.8	B
Near Bridge Street	WB	150	0.18	15.4	B

APPENDIX C — PROPOSED CONDITION CAPACITY ANALYSIS OUTPUT

Per-intersection HCM 6 solver output for the Proposed Build condition (No-Build volumes plus project external trips at the assigned distribution).

Intersection	Approach	Vol (vph)	v/c	Delay (s)	LOS
Near Erie Boulevard East	NB	605	0.75	26.7	C
Near Erie Boulevard East	SB	497	0.61	22.3	C
Near Erie Boulevard East	EB	476	0.59	21.6	C
Near Erie Boulevard East	WB	409	0.51	19.9	B
Thompson Road & Erie Boulevard	NB	193	0.24	15.9	B
Thompson Road & Erie Boulevard	SB	171	0.21	15.6	B
Thompson Road & Erie Boulevard	EB	132	0.16	15.1	B
Thompson Road & Erie Boulevard	WB	120	0.15	15.0	B
Erie Boulevard East & Bridge Street	NB	647	0.80	29.3	C
Erie Boulevard East & Bridge Street	SB	412	0.51	19.9	B
Erie Boulevard East & Bridge Street	EB	527	0.65	23.3	C
Erie Boulevard East & Bridge Street	WB	379	0.47	19.2	B
Near Bridge Street	NB	226	0.28	16.4	B
Near Bridge Street	SB	190	0.23	15.9	B
Near Bridge Street	EB	186	0.23	15.8	B
Near Bridge Street	WB	153	0.19	15.4	B
Near Erie Boulevard East	NB	275	0.34	17.2	B
Near Erie Boulevard East	SB	215	0.27	16.3	B
Near Erie Boulevard East	EB	205	0.25	16.1	B
Near Erie Boulevard East	WB	169	0.21	15.6	B
Near Bridge Street	NB	193	0.24	15.9	B
Near Bridge Street	SB	186	0.23	15.8	B
Near Bridge Street	EB	196	0.24	16.0	B
Near Bridge Street	WB	165	0.20	15.6	B
Near Erie Boulevard East	NB	31	0.04	13.9	B
Near Erie Boulevard East	SB	29	0.04	13.9	B
Near Erie Boulevard East	EB	29	0.04	13.9	B
Near Erie Boulevard East	WB	31	0.04	13.9	B
Near East Genesee Street	NB	34	0.04	14.0	B
Near East Genesee Street	SB	32	0.04	14.0	B
Near East Genesee Street	EB	25	0.03	13.9	B
Near East Genesee Street	WB	24	0.03	13.9	B
Near Bridge Street	NB	195	0.24	16.0	B
Near Bridge Street	SB	189	0.23	15.9	B
Near Bridge Street	EB	193	0.24	15.9	B

Intersection	Approach	Vol (vph)	v/c	Delay (s)	LOS
Near Bridge Street	WB	154	0.19	15.4	B

APPENDIX D — CRASH ANALYSIS DIAGRAMS AND TABLES

Crash analysis diagrams and tables are not produced by this screening analysis. When the §4.0 crash-analysis section is expanded for formal submittal, the following NYSDOT forms are required: TE-156a (Collision Diagram), TE-164a (Safety Benefits Evaluation Form), TE-204a (Accident Rate Summary), and TE-213 (HAL / PIL / SDL Screening Output). SIMS output and statewide-average rate references should be obtained from the controlling Regional Traffic Office per HDM Chapter 5 §5.3.4 and the NYSDOT Office of Modal Safety statewide rate tables (<https://www.dot.ny.gov/divisions/operating/osss/highway/accident-rates>).

This report is based on the NYSDOT TIS Shell revised on 9/16/2014.

FOUR-STEP TRAVEL DEMAND MODEL

Off-site project trips are distributed and assigned with the four-step urban transportation modeling process (FHWA; NCHRP Report 716, Travel Demand Forecasting). Each step below shows what the engine computed for this site.

Step 1 — Trip Generation

Public-data screening trip rates (SANDAG 2002, corroborated by NHTS 2017 / NCHRP 716) applied to the proposed land use give the site's productions.

Land use	LU 820 — Shopping Center / Retail
Size	85 1,000 sqft GLA
Daily trips (productions)	6,800
PM peak-hour trips	680 (326 in / 354 out)

Step 2 — Trip Distribution (Gravity Model)

A production-constrained gravity model allocates trips to surrounding signals: $T_j = P \cdot (A_j \cdot F_j) / \sum (A_x \cdot F_x)$. Attractiveness A_j is each signal's through-volume; the friction factor F_j is the NCHRP-716 gamma function $F = a \cdot t^b \cdot e^{(c \cdot t)}$ (home-based-work: $a=28507$, $b=-0.02$, $c=-0.123$) on the travel time t to the signal.

Signal (attraction zone)	Dist mi	Time min	PM trips	Share
Near Erie Boulevard East	0.35	0.8	77	23.9%
Thompson Road & Erie Boulevard East	0.41	1.0	59	18.3%
Erie Boulevard East & Bridge Street	0.42	1.0	54	16.8%
Near Bridge Street	0.50	1.2	36	11.2%
Near Erie Boulevard East	0.54	1.3	31	9.6%
Near Bridge Street	0.61	1.5	21	6.5%
Near Erie Boulevard East	0.64	1.5	18	5.6%
Near East Genesee Street	0.71	1.7	13	4.0%
Near Bridge Street	0.71	1.7	13	4.0%

Top 12 distribution zones by assigned PM-peak trips shown; the full set is in the capacity appendix.

Step 3 — Mode Choice

Auto-mode share 94% applied via a binary logit $P(\text{auto}) = 1 / (1 + e^{-(ASC - \lambda \cdot \Delta GC)})$ calibrated to the metro's measured auto-mode share (ACS B08301) and shifted by site urbanity (local density), so a denser, more transit-served site splits further from auto than a greenfield parcel in the same metro. The remaining 6% of trips (transit, walking, cycling) do not load the off-site roadway network; only auto trips are carried into Step 4.

Step 4 — Route Assignment

A capacity-constrained assignment using the BPR volume-delay function $t = t_0 \cdot [1 + 0.15 \cdot (v/c)^4]$ iteratively shifts trips away from over-capacity signals toward less-congested alternatives until the assignment is congestion-consistent. Highest post-build v/c among study signals: 1.10. Per-signal v/c, delay, LOS, and queue are in the capacity appendix. (Road-network corridor routing was not available for this region; per-signal capacity-constrained assignment was used.)

APPENDIX — INTERSECTION CAPACITY ANALYSIS WORKSHEETS

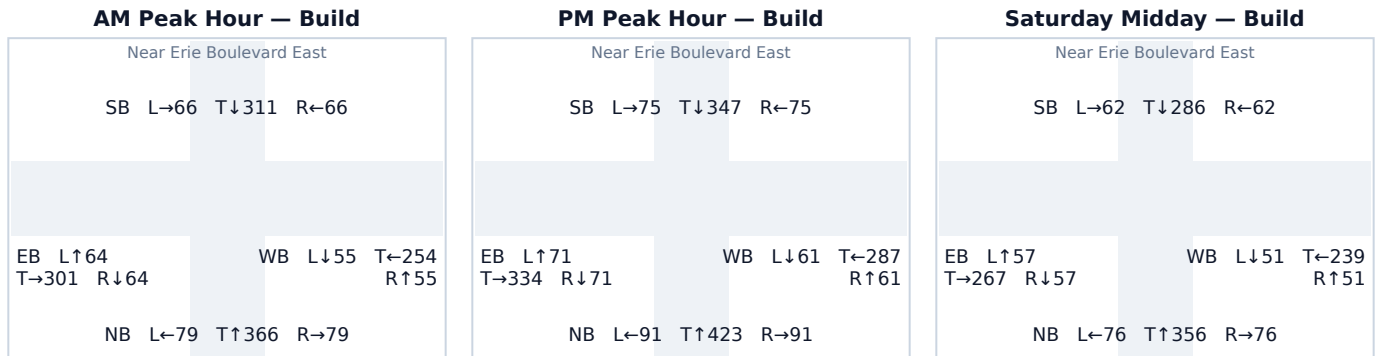
One worksheet per study intersection: a turning-movement diagram for each analyzed peak period plus the per-approach HCM capacity table. Analysis follows the Highway Capacity Manual 6th Edition signalized-intersection method; v/c, control delay (sec/veh), Level of Service, and 95th-percentile back-of-queue (ft) are reported for the Existing (No-Build) and Build conditions.

Background turning-movement volumes in the diagrams are distributed from each approach total using an estimated 15/70/15 (Left/Through/Right) split. Project-trip movements are assigned geometrically from the study's directional trip distribution (see each worksheet's Affected movements table). Replace both with measured turning-movement counts (TMCs) before a formal submittal.

A.1 Near Erie Boulevard East

Signal ID	syracuse-999
Location	Central Syracuse · 0.35 mi from site
Added PM peak trips	77
Existing / No-Build (PM)	LOS E · 73.7 s/veh · v/c 1.06
Build (PM)	LOS F · 88.6 s/veh · v/c 1.10
Δ control delay	+14.9 s/veh (LOS change)
Mitigation	Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s.

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	Exist vph	+Trips	Build vph	v/c Ex→Bld	Delay Ex→Bld	LOS Ex→Bld	Q95 ft
NB	565	40	605	0.70 → 0.75	24.8 → 26.7	C → C	517
SB	479	18	497	0.59 → 0.61	21.7 → 22.3	C → C	389
EB	476	0	476	0.59 → 0.59	21.6 → 21.6	C → C	367
WB	390	19	409	0.48 → 0.51	19.4 → 19.9	B → B	301

Affected movements (PM peak project trips)

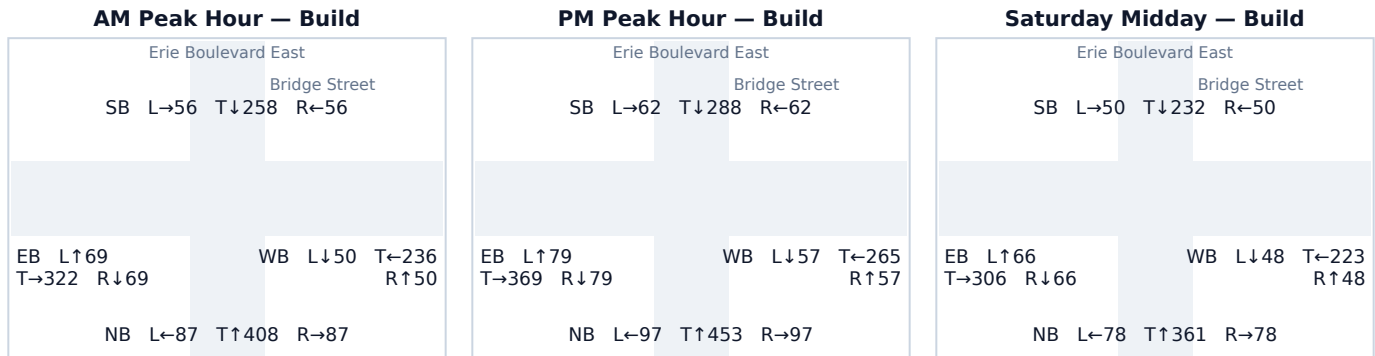
Movement	Project trips	% of project trips
NB Right	21	27%
WB Left	19	25%
NB Through	19	25%
SB Through	18	23%

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.2 Erie Boulevard East & Bridge Street

Signal ID	syracuse-592
Location	Central Syracuse · 0.42 mi from site
Added PM peak trips	54
Existing / No-Build (PM)	LOS E · 73.7 s/veh · v/c 1.06
Build (PM)	LOS F · 83.9 s/veh · v/c 1.09
Δ control delay	+10.2 s/veh (LOS change)
Mitigation	Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s.

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	Exist vph	+Trips	Build vph	v/c Ex→Bld	Delay Ex→Bld	LOS Ex→Bld	Q95 ft
NB	647	0	647	0.80 → 0.80	29.3 → 29.3	C → C	572
SB	408	3	412	0.50 → 0.51	19.8 → 19.9	B → B	303
EB	499	28	527	0.62 → 0.65	22.3 → 23.3	C → C	423
WB	357	23	379	0.44 → 0.47	18.7 → 19.2	B → B	272

Affected movements (PM peak project trips)

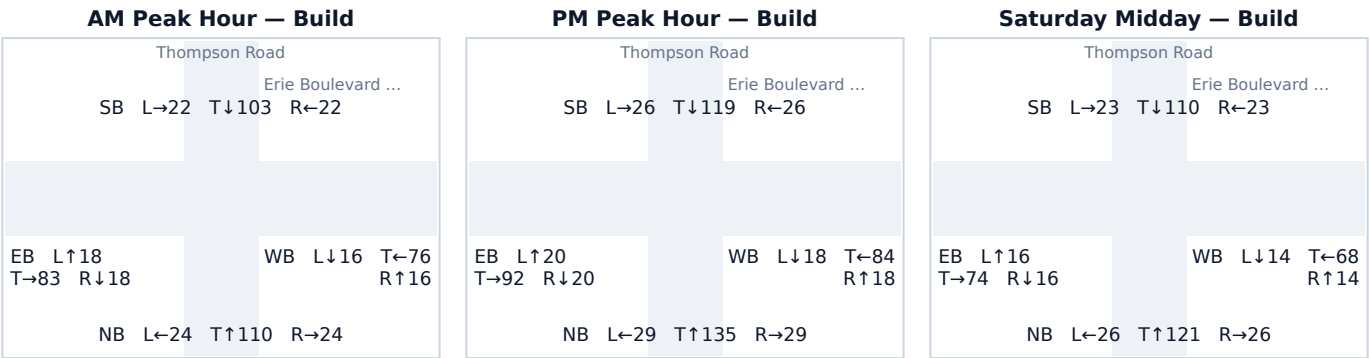
Movement	Project trips	% of project trips
EB Through	24	44%
WB Through	23	43%
EB Left	4	7%
SB Right	3	6%

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.3 Thompson Road & Erie Boulevard East

Signal ID	syracuse-774
Location	Central Syracuse · 0.41 mi from site
Added PM peak trips	59
Existing / No-Build (PM)	LOS B · 16.8 s/veh · v/c 0.31
Build (PM)	LOS B · 17.2 s/veh · v/c 0.34
Δ control delay	+0.4 s/veh
Mitigation	No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	Exist vph	+Trips	Build vph	v/c Ex→Bld	Delay Ex→Bld	LOS Ex→Bld	Q95 ft
NB	162	31	193	0.20 → 0.24	15.5 → 15.9	B → B	123
SB	143	27	171	0.18 → 0.21	15.3 → 15.6	B → B	107
EB	131	1	132	0.16 → 0.16	15.1 → 15.1	B → B	81
WB	120	0	120	0.15 → 0.15	15.0 → 15.0	B → B	73

Affected movements (PM peak project trips)

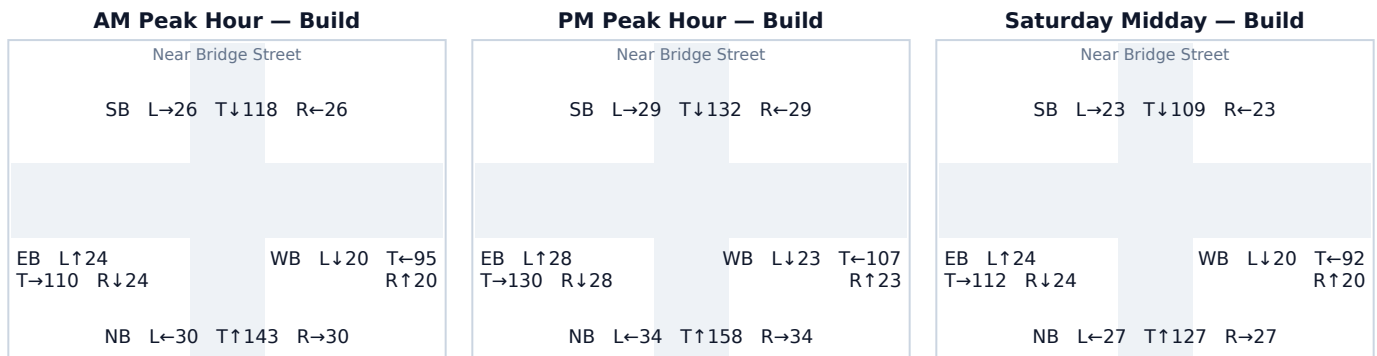
Movement	Project trips	% of project trips
NB Through	30	51%
SB Through	27	46%
NB Left	1	2%
EB Right	1	2%

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.4 Near Bridge Street

Signal ID	syracuse-238
Location	Central Syracuse · 0.50 mi from site
Added PM peak trips	36
Existing / No-Build (PM)	LOS B · 18.1 s/veh · v/c 0.40
Build (PM)	LOS B · 18.4 s/veh · v/c 0.42
Δ control delay	+0.3 s/veh
Mitigation	No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	Exist vph	+Trips	Build vph	v/c Ex→Bld	Delay Ex→Bld	LOS Ex→Bld	Q95 ft
NB	226	0	226	0.28 → 0.28	16.4 → 16.4	B → B	147
SB	186	3	190	0.23 → 0.23	15.8 → 15.9	B → B	121
EB	167	19	186	0.21 → 0.23	15.6 → 15.8	B → B	117
WB	140	14	153	0.17 → 0.19	15.2 → 15.4	B → B	95

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
EB Through	15	42%
WB Through	14	39%
EB Left	4	11%
SB Right	3	8%

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.5 Near Erie Boulevard East

Signal ID	syracuse-588
Location	Central Syracuse · 0.54 mi from site
Added PM peak trips	31
Existing / No-Build (PM)	LOS B · 19.1 s/veh · v/c 0.46
Build (PM)	LOS B · 19.4 s/veh · v/c 0.48
Δ control delay	+0.3 s/veh
Mitigation	No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)

AM Peak Hour — Build	PM Peak Hour — Build	Saturday MIDDAY — Build
Near Erie Boulevard East SB L→29 T↓136 R←29 EB L↑27 T→123 R↓27 WB L↓22 T←105 R↑22 NB L←37 T↑173 R→37	Near Erie Boulevard East SB L→32 T↓151 R←32 EB L↑31 T→143 R↓31 WB L↓25 T←119 R↑25 NB L←41 T↑193 R→41	Near Erie Boulevard East SB L→26 T↓120 R←26 EB L↑26 T→122 R↓26 WB L↓22 T←102 R↑22 NB L←33 T↑154 R→33

Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	Exist vph	+Trips	Build vph	v/c Ex→Bld	Delay Ex→Bld	LOS Ex→Bld	Q95 ft
NB	275	0	275	0.34 → 0.34	17.2 → 17.2	B → B	184
SB	215	0	215	0.27 → 0.27	16.3 → 16.3	B → B	139
EB	189	16	205	0.23 → 0.25	15.9 → 16.1	B → B	131
WB	155	15	169	0.19 → 0.21	15.4 → 15.6	B → B	106

Affected movements (PM peak project trips)

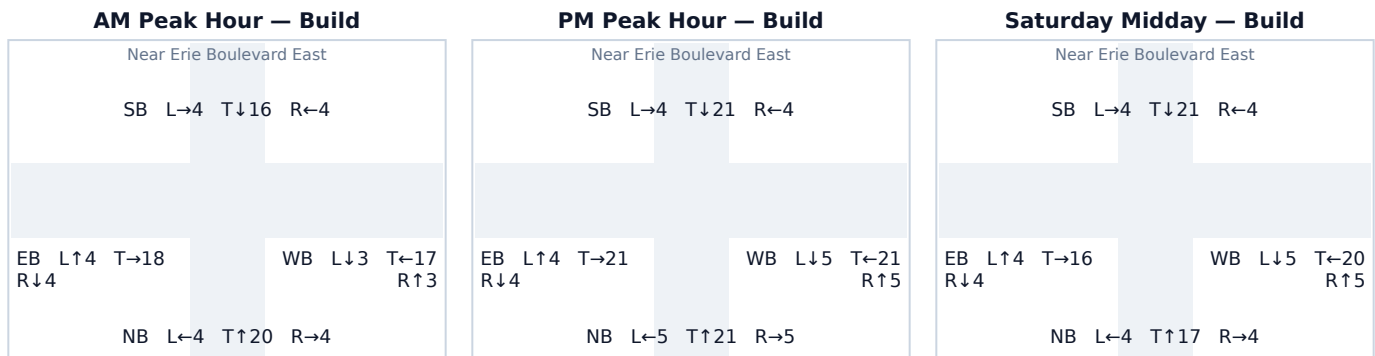
Movement	Project trips	% of project trips
EB Through	16	52%
WB Through	15	48%

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.6 Near Erie Boulevard East

Signal ID	syracuse-595
Location	Central Syracuse · 0.64 mi from site
Added PM peak trips	18
Existing / No-Build (PM)	LOS B · 14.1 s/veh · v/c 0.06
Build (PM)	LOS B · 14.2 s/veh · v/c 0.07
Δ control delay	+0.1 s/veh
Mitigation	No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	Exist vph	+Trips	Build vph	v/c Ex→Bld	Delay Ex→Bld	LOS Ex→Bld	Q95 ft
NB	31	0	31	0.04 → 0.04	13.9 → 13.9	B → B	18
SB	21	8	29	0.03 → 0.04	13.8 → 13.9	B → B	17
EB	28	1	29	0.03 → 0.04	13.9 → 13.9	B → B	17
WB	22	9	31	0.03 → 0.04	13.8 → 13.9	B → B	18

Affected movements (PM peak project trips)

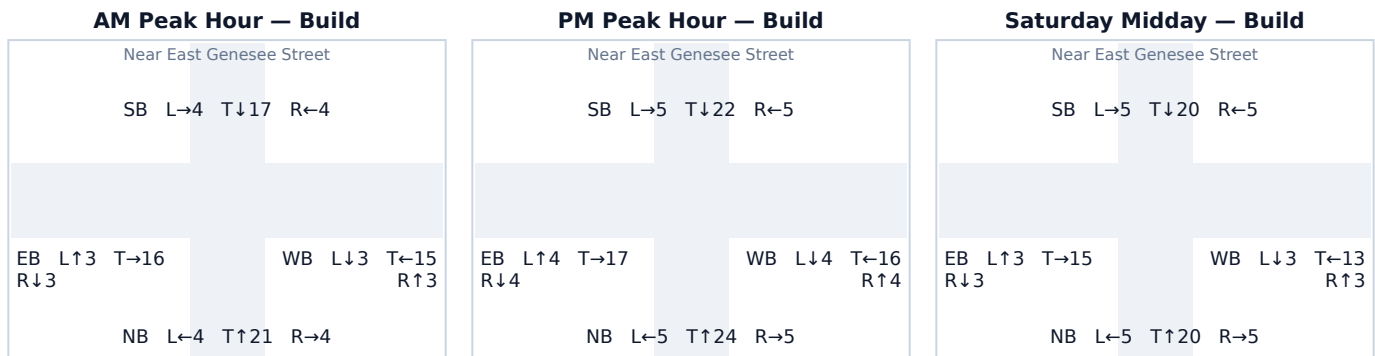
Movement	Project trips	% of project trips
WB Right	8	44%
SB Left	8	44%
WB Through	1	6%
EB Through	1	6%

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.7 Near East Genesee Street

Signal ID	syracuse-74
Location	Central Syracuse · 0.71 mi from site
Added PM peak trips	13
Existing / No-Build (PM)	LOS B · 14.1 s/veh · v/c 0.06
Build (PM)	LOS B · 14.2 s/veh · v/c 0.06
Δ control delay	+0.1 s/veh
Mitigation	No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	Exist vph	+Trips	Build vph	v/c Ex→Bld	Delay Ex→Bld	LOS Ex→Bld	Q95 ft
NB	29	5	34	0.04 → 0.04	13.9 → 14.0	B → B	19
SB	25	7	32	0.03 → 0.04	13.9 → 14.0	B → B	19
EB	24	1	25	0.03 → 0.03	13.9 → 13.9	B → B	15
WB	24	0	24	0.03 → 0.03	13.9 → 13.9	B → B	14

Affected movements (PM peak project trips)

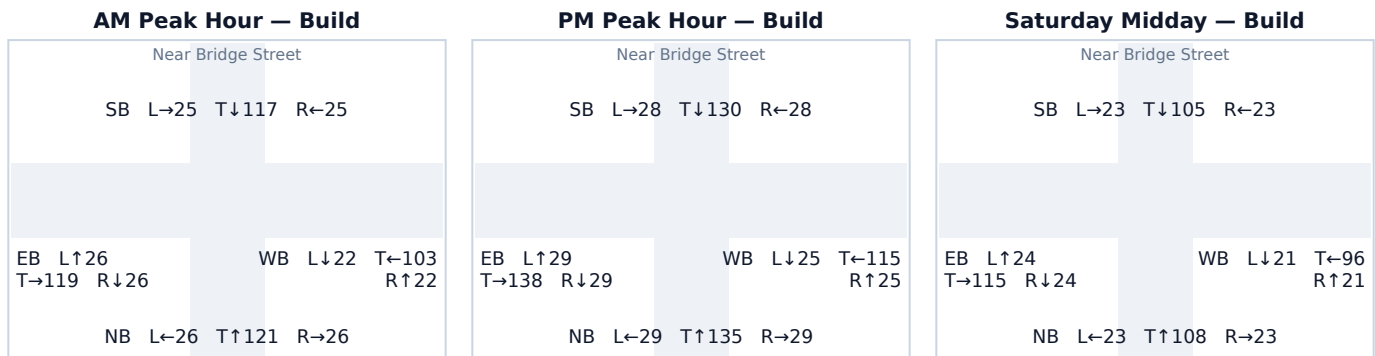
Movement	Project trips	% of project trips
SB Through	5	38%
NB Through	5	38%
SB Right	2	15%
EB Left	1	8%

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.8 Near Bridge Street

Signal ID	syracuse-924
Location	Central Syracuse · 0.61 mi from site
Added PM peak trips	21
Existing / No-Build (PM)	LOS B · 18.1 s/veh · v/c 0.40
Build (PM)	LOS B · 18.2 s/veh · v/c 0.41
Δ control delay	+0.1 s/veh
Mitigation	No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	Exist vph	+Trips	Build vph	v/c Ex→Bld	Delay Ex→Bld	LOS Ex→Bld	Q95 ft
NB	193	0	193	0.24 → 0.24	15.9 → 15.9	B → B	122
SB	184	2	186	0.23 → 0.23	15.8 → 15.8	B → B	118
EB	185	11	196	0.23 → 0.24	15.8 → 16.0	B → B	125
WB	158	8	165	0.19 → 0.20	15.5 → 15.6	B → B	103

Affected movements (PM peak project trips)

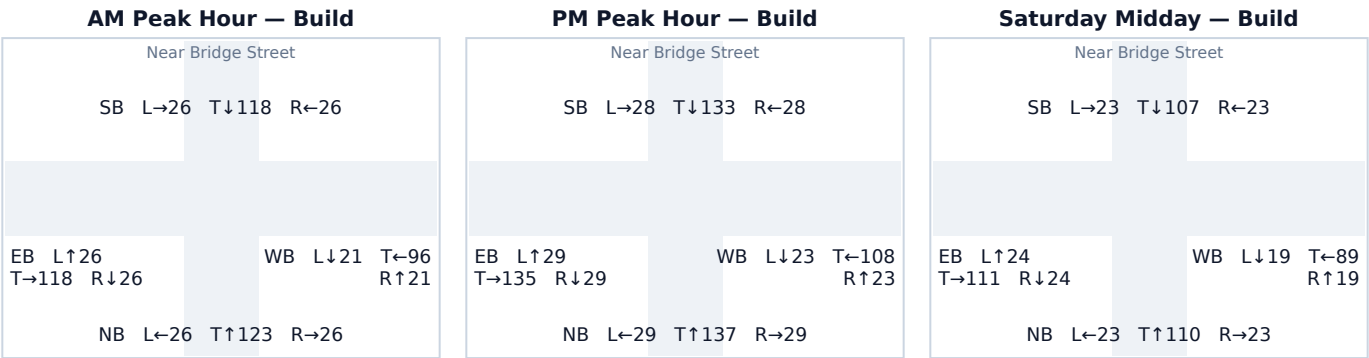
Movement	Project trips	% of project trips
EB Through	8	38%
WB Through	8	38%
EB Left	3	14%
SB Right	2	10%

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.9 Near Bridge Street

Signal ID	syracuse-920
Location	Central Syracuse · 0.71 mi from site
Added PM peak trips	13
Existing / No-Build (PM)	LOS B · 18.1 s/veh · v/c 0.40
Build (PM)	LOS B · 18.2 s/veh · v/c 0.41
Δ control delay	+0.1 s/veh
Mitigation	No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	Exist vph	+Trips	Build vph	v/c Ex→Bld	Delay Ex→Bld	LOS Ex→Bld	Q95 ft
NB	195	0	195	0.24 → 0.24	16.0 → 16.0	B → B	124
SB	188	2	189	0.23 → 0.23	15.9 → 15.9	B → B	120
EB	186	7	193	0.23 → 0.24	15.8 → 15.9	B → B	123
WB	150	4	154	0.18 → 0.19	15.4 → 15.4	B → B	96

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
EB Through	5	38%
WB Through	4	31%
EB Left	2	15%
SB Right	2	15%

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.